



Blood Bank & Donor Locator App

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Guide :

Department :

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Overview :

- A mobile application designed to quickly locate **blood donors and nearby blood**
- Helps patients and their families during **medical emergencies**.
- Donors can register their **blood group, contact details and location**.
- Users can search for suitable donors and nearby blood banks.
- Location-based search helps identify nearby available resources.

Technologies

React Native

JavaScript

Node.js & Express.js

Firebase Firestore

Firebase Authentication

Google Maps API

Expected Outcome

A user-friendly system that improves **blood accessibility and donor coordination.**

These points are directly based on your submitted abstract.

Keywords:

Blood Donation | Donor Locator | Blood Bank | Location-Based Search

Problem Statement

Existing Problem

During medical emergencies, finding suitable blood donors or nearby blood banks can take a lot of time.

Who is affected?

- Patients
- Patient families
- Hospitals
- Blood banks
- Blood donors

Why solve it?

- Blood may be required immediately.
- Searching through multiple sources is time-consuming.
- It can be difficult to find a suitable blood group quickly.
- Contacting donors/blood banks may take additional time.

Limitations of Existing Approach :

- Information may be scattered across different sources.
- No single centralized platform.
- Difficult to identify nearby donors quickly.
- Manual communication can cause delays.

Proposed Problem Definition :

To develop a centralized mobile application that helps users **find suitable blood donors and nearby blood banks quickly using blood-group and location-based search.**

Motivation

Why did we select this project?

- Blood is often required in **emergency situations**.
- Finding a compatible donor quickly can be difficult.
- Existing information may not be available in one place.
- Technology can make donor coordination faster and easier.

Real-World Importance

The application can help reduce the **time and effort required to locate blood**.

Expected Benefits

- Faster donor search
- Nearby blood bank identification
- Better donor-recipient communication
- Organized donor records
- Support for emergency blood requests
- Encouragement of voluntary blood donation

Objectives

Objective 1

To provide a centralized platform for registering and managing blood donor information.

Objective 2

To help users find suitable donors based on **blood group and location**.

Objective 3

To help users locate nearby blood banks using **location-based services and maps**.

Objective 4

To improve communication between **donors, recipients, hospitals and blood banks**.

Objective 5

To reduce the time and effort required to find blood during emergencies.

Literature Servey

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Author & Year	Method/Technique	Dataset	Major Finding	Limitation
Study 1	Blood donor management	Donor records	Helps manage donor information	Limited location support
Study 2	Location-based service	Location data	Helps locate nearby services	Limited donor coordination
Study 3	Mobile blood bank system	Blood bank data	Improves accessibility	Limited centralized features

Research Gap

Limitations in Existing Systems

- Donor information may not be available in one centralized platform.
- Users may have difficulty finding **nearby suitable donors**.
- Blood bank information and donor information may be separated.
- Emergency communication can still involve manual effort.
- Location-based donor and blood bank coordination can be improved.

Research Gap

Existing systems mainly focus on blood bank management or donor registration separately. There is a need for an integrated platform that combines donor information, blood bank information, blood-group search and location-based services. Our proposed application addresses this gap by providing these features through a single mobile platform.

Proposed Methodology

User Registration / Login



Enter Blood Requirement



Select Blood Group



Location-Based Search



Find Suitable Donors / Nearby Blood Banks



Retrieve Information from Firebase



Display Results on Mobile App



Contact / Emergency Request

Explain:

Input:

Blood group, location and user requirement.

Preprocessing:

Validate user information and search criteria.

Processing:

Search donor and blood bank records using blood group and location.

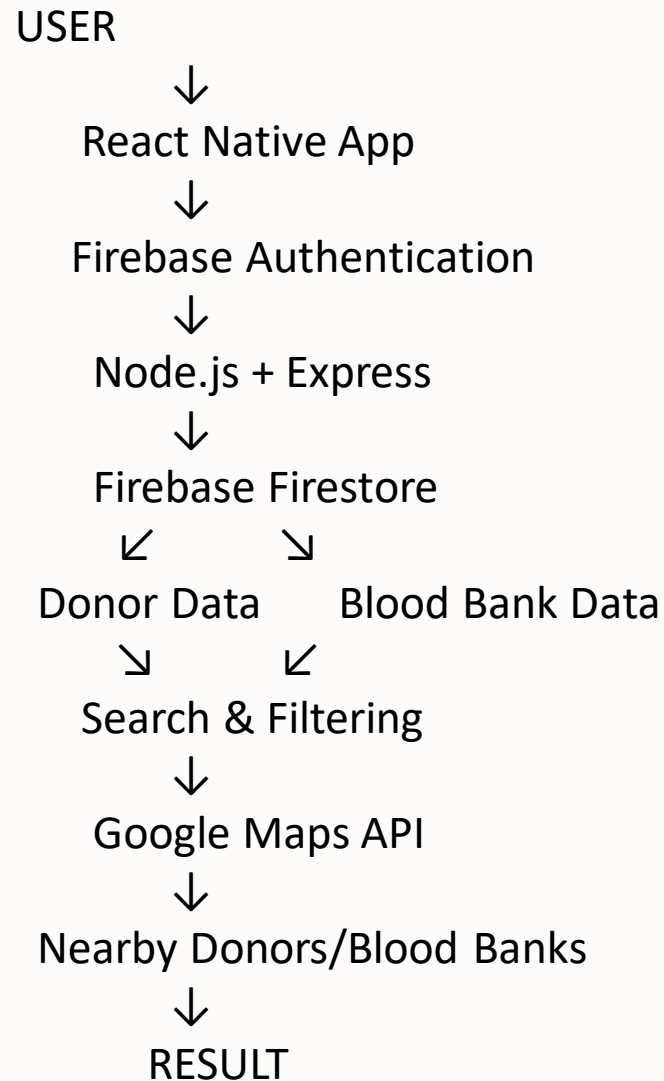
Output:

List of suitable donors and nearby blood banks.

Performance Evaluation:

Check response time, search accuracy and usability

System Architecture



Main Components

1.User Interface – React Native

2.Authentication – Firebase

Authentication

3.Backend – Node.js + Express.js

4.Database – Firebase Firestore

5.Location Services – Google Maps API

Technologies Used

Technology	Purpose
React Native	Mobile application frontend
JavaScript	Application programming
Node.js	Backend runtime
Express.js	Backend/API development
Firebase Firestore	Store donor and blood bank data
Firebase Authentication	User authentication
Google Maps API	Location and nearby search
Visual Studio Code	Development
Android Studio	Android development
Postman	API testing
Git & GitHub	Version control
Firebase	Cloud platform

Input Details

- User name and contact details
- Blood group
- User location
- Blood requirement
- Donor information
- Blood bank information
- Search criteria such as blood group and location

Input: Blood group + location + user requirement.

Output: Suitable donors and nearby blood banks.

User Registration & Authentication

- User registers in the application.
- User provides basic details and contact information.
- Blood donors can enter their blood group and location.
- Firebase Authentication is used for secure user authentication.
- Donor information is stored in the database.

Main Technologies: React Native + Firebase Authentication + Firebase Firestore.

Blood Donor Search

- User selects the required blood group.
- User provides or uses their location.
- System searches for suitable donors.
- Donor records are retrieved from Firebase Firestore.
- Results are filtered using blood group and location.
- Suitable donors are displayed in the mobile application.

Goal: Find compatible donors quickly

Blood Bank & Location Module

- User searches for nearby blood banks.
- Google Maps API is used for location-based services.
- Blood bank information is retrieved from the system.
- Nearby blood banks are displayed to the user.
- Users can access essential blood bank information quickly.

Main Technology: Google Maps API.

Experimental Setup

Development Environment

- **Frontend:** React Native
- **Programming Language:** JavaScript
- **Backend:** Node.js + Express.js
- **Database:** Firebase Firestore
- **Authentication:** Firebase Authentication
- **Maps:** Google Maps API
- **IDE:** Visual Studio Code
- **Android Development:** Android Studio
- **API Testing:** Postman
- **Version Control:** Git & GitHub

Testing Areas

- User authentication
- Donor search
- Blood group filtering
- Location-based search
- Blood bank search

Expected Results

Users can register and access the application.

Donor information can be stored and managed.

Users can search donors based on blood group.

Nearby blood banks can be identified.

Location-based search helps find nearby resources.

The system can reduce the time and effort needed to locate blood.

Expected Outcome: A user-friendly system for better blood accessibility and donor coordination.

Note: If your application is already working, you can replace “Expected Results” with your actual results.

Performance Evaluation

Evaluation Parameters

We evaluate the system based on:

- **Response Time** – How quickly search results are displayed.
- **Search Accuracy** – Whether suitable donors/blood banks are identified.
- **Usability** – How easy the application is to use.
- **Location Accuracy** – How effectively nearby resources are identified.
- **System Reliability** – Whether the main functions work correctly.

These evaluation parameters are included in your project methodology.

Screenshots / Working Demo



Comparison with Existing System

Existing System	Proposed System
Information may be scattered	Centralized platform
Difficult to find nearby donors	Location-based donor search
Blood bank and donor information may be separate	Both can be accessed through one application
Manual communication may cause delays	Faster access to information
Limited location support	Google Maps-based location services

Challenges & Solutions

Challenge	Solution
Finding suitable blood donors	Blood-group based search
Finding nearby donors	Location-based search
Scattered information	Centralized database
Finding nearby blood banks	Google Maps API
Managing donor information	Firebase Firestore
User authentication	Firebase Authentication
Communication delays	Quick access to donor/blood bank information

Future Improvements

- Improve location-based donor matching.
- Add more emergency request features.
- Improve communication between donors and recipients.
- Expand blood bank information.
- Improve search and filtering features.
- Make the application more accessible and user-friendly.
- Support better coordination between hospitals, donors and blood banks.

Conclusion

- The Blood Bank & Donor Locator App provides a centralized platform for blood-related information.
- Users can search for suitable donors using blood group and location.
- The application helps users find nearby blood banks.
- Firebase provides authentication and data storage.
- Google Maps API supports location-based services.
- The system aims to reduce the time and effort required to locate blood during emergencies.

References

- [Firebase Documentation](#)
- [React Native Documentation](#)
- [Node.js Documentation](#)
- [Express.js Documentation](#)
- [Google Maps Platform Documentation](#)
- [Android Studio Documentation](#)
- [Postman Documentation](#)
- [Git & GitHub Documentation](#)



Thank you